

PFM 206 Course Outline

PFM 206 — Budget Output Engine

Course Outline

School	02 — Macroeconomics and Public Finance
Level	200 (analyst / practitioner)
Hours	30 (indicative band 24–32)
Shape	5 modules · 10 sessions · 6 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101. PFM 204 and PFM 205 are strongly recommended and are effectively assumed: this engine reads the budget PFM 204 produces and resolves against the parameter service PFM 205 teaches, and its central finding is PFM 205's limitation L-01 arriving in a conversion. PFM 203 is useful for the ceiling and recurrent-baseline handovers. No Julia is required: no laboratory needs a running instance.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-24 · 12-month cycle

What this course is

PFM 206 teaches the AICTEM Budget Output Engine, version 2: a 6,861-line Julia implementation across 4 layers that converts a government budget into physical resources, transactional capacity, institutional capacity, service outputs, future recurrent obligations and model-ready fiscal and sector shocks. It carries 73 registered indicators across 17 sections, 1,413 test assertions in 5 suites, and 17 documents of its own.

The engine answers six questions about a budget and refuses a seventh. It runs 15 stages in one transaction, converts in four modes declared by a crosswalk rather than inferred from a sector, reconciles at five levels along five dimensions, and reports everything it could not convert in a closed vocabulary of diagnostics that each carry the amount of money they concern.

The Budget Output Engine is where the arithmetic happens. The parameter service (PFM 205) holds the coefficients. The Budget module (PFM 204) holds the classified budget. The MTEF (PFM 203) holds the ceilings, and FINEX (MAC 202) the macro frame.

Who it is for

Budget and planning analysts asked what a budget delivers rather than what it contains; sector planning officers who receive a converted plan; monitoring and evaluation staff reconciling spending with output; public investment officers costing the recurrent consequence of a capital decision; and EDIS developers maintaining the engine or any module that consumes its output.

What this course does not claim

This course does not produce outcomes. GDP, poverty, mortality and learning belong to the sector, epidemiological, econometric, CGE and microsimulation models, and the parameter service refuses Layer C coefficients to the conversion path so the engine cannot compute them even by accident.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	What the engine is	01	01	3.2
M2	The mechanics of a conversion	02–04	02, 04	9.6
M3	Beyond the conversion	05–07	01, 04–05	5.1
M4	The utopia pilot	08–09	03, 05	7.9
M5	The indicator library	10	06	3.2
ASM	Assessment and revision	—	—	1.0
	Total	10	6	30.0

The 10 sessions and 6 laboratories, in teaching order, and the subject matter of each:

M1 What the engine is — Session 01 · 3.2 h · Laboratory 01

- **01 What the engine is, and what it will not do** — the question a budget cannot answer; six questions, and the seventh it refuses; where the engine sits; ten things the engine will not do, and more
- *Lab 01 Read the run* — establish for yourself what the pilot run did: what it read, what it converted, what it refused, and that the identity closes

M2 The mechanics of a conversion — Sessions 02–04 · 9.6 h · Laboratory 02, 04

- **02 The crosswalk and the four modes** — the crosswalk declares the mode; the physical mode; the transactional mode; the institutional mode, and more
- *Lab 02 Convert a line in each mode* — perform one conversion in each of the four modes by hand, and report each under the eight-element contract
- **03 Stocks, capacity, output and the run** — stocks, capacity, output and the bottleneck; the run: thirteen stages in one transaction
- **04 Capital, commissioning and execution** — capital stock, commissioning and depreciation; execution: two paths, and which one was used
- *Lab 04 Trace a diagnostic to its money* — take each of the four diagnostics and establish exactly what it means, what it costs, and who fixes it

M3 Beyond the conversion — Sessions 05–07 · 5.1 h · Laboratory 01, 04–05

- **05 Recurrent obligations, lifecycle and uncertainty** — the capital-to-recurrent engine; lifecycle cost; uncertainty: cases run as complete conversions
- *Lab 05 Build the acquisition programme* — turn the pilot's ten per cent conversion rate into a costed, sequenced data-acquisition plan
- **06 Indicators, resolution and reconciliation** — the indicator registry; coefficient resolution, and what a refusal carries; reconciliation: five levels, five dimensions
- **07 Diagnostics and the handovers** — diagnostics: the §44 vocabulary; MTEF and FINEX; CDCGE; MEL and EDIS, and more

M4 The utopia pilot — Sessions 08–09 · 7.9 h · Laboratory 03, 05

- **08 The Utopia pilot I — inputs and coverage** — the run, and what it read; three coverage questions, three answers; the outcome, in three parts
- *Lab 03 Audit the crosswalk* — establish where the pilot's coverage gap is, and write the rules that would close the largest part of it
- **09 The Utopia pilot II — diagnostics and the gap** — the diagnostics the run raised; what was not validated, and why that is the report; what Utopia must measure next; the acquisition programme, ranked by money

M5 The indicator library — Session 10 · 3.2 h · Laboratory 06

- **10 The library, ten errors and reporting honestly** — how to read the library; health and education; agriculture, transport and the appraisal arithmetic; energy, water and sanitation, and more
- *Lab 06 Report a conversion honestly* — write the results section a ministry could publish, and the one it could not

What a graduate can do

1. Explain what a budget cannot say about delivery, and what a conversion engine adds.
2. State the six questions the engine answers and the seventh it refuses, and say what enforces the refusal.
3. Place the engine in the production chain and name what each link owns.
4. Name the ten standing refusals and the failure each prevents.
5. Explain why the conversion mode is declared by the crosswalk rather than inferred from the sector.
6. Perform a conversion in each of the four modes, and name the binding capacity in the physical case.
7. Distinguish stocks, capacities, outputs and bottlenecks, and say why counts and values are held separately.
8. Name the thirteen pipeline stages and give three reasons the order is not arbitrary.
9. State what lets a capital investment enter the stock, and why $= 0$ is a claim rather than a default.
10. Distinguish declared from measured execution, and say what each makes a result.

The 17 formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	89 items across 10 modules
Laboratory exercises	30 %	6 laboratories, equally weighted, marked on method, interpretation, honesty about limits and communication
Applied assignments	25 %	read the run and verify the identity; convert a line in each of the four modes; audit the crosswalk and the diagnostics; build the acquisition programme and report it
Final assessment	30 %	Three hours, open book: definitions and structure; calculation; interpretation; reporting

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been recorded with its actual value: an unchecked result is no result.

Materials and set-up

A 150-page technical manual; 10 lecture decks and 6 laboratory decks; a laboratory manual, instructor guide and learner workbook; a 89-item quiz bank, 4 assignments and a final assessment with its marking guide; 8 datasets; 4 code exercises; 5 case studies; and the lms package. The README lists every file.

Every figure in this course is traceable to a published source or a committed model output; the sources are registered in `COURSE_VERSION.json` and the limitations in `COURSE_BUILD_REPORT.md`. Laboratories need only the extracts shipped in `08_Data/`.

PFM 206 Budget Output Engine · version 1.0 · AICTEM EDIS Academy, School 02 Macroeconomics and Public Finance.